

Equations with Brackets

AQA · KS3 Year 8

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

$3(x + 2) = 3x + 6$, not $3x + 2$.

A negative outside flips both signs inside.

Do not divide before expanding when other terms are present.

Watch signs when subtracting a bracket, e.g. $-2(x - 4) = -2x + 8$.

METHOD

- 1 Expand every bracket fully
- 2 Collect like terms on each side
- 3 Move the smaller letter term across
- 4 Collect numbers, then divide
- 5 Check in the original equation

WORKED EXAMPLE

Solve $3(x + 2) = 18$.

Expand: $3x + 6 = 18$. Subtract 6: $3x = 12$. Divide by 3: $x = 4$.

Check: $3(4 + 2) = 18$

VISUAL AID

Area model: a rectangle split into 3 lots of $(x + 2)$. The total area $3x + 6$ equals 18, so each $x = 4$.

1. (2 marks)

Solve: a) $2(x + 3) = 14$ b) $5(x - 1) = 20$

2. (2 marks)

Solve: a) $3(x + 4) = 18$ b) $4(2x - 1) = 28$

3. (3 marks)

Solve: a) $2(3x + 1) = 26$ b) $5(2x - 3) = 35$ c) $3(x + 5) = 12$

4. (3 marks)

Solve: a) $2(x + 4) = x + 11$ b) $4(x - 1) = 2(x + 3)$ c) $3(2x + 1) = 4x + 11$

5. (3 marks)

Solve: a) $2(x + 1) + 3(x - 2) = 11$ b) $5(x - 1) - 2(x - 4) = 18$ c) $4(x + 2) - 3(x - 1) = 15$

6. (3 marks)

Rectangle length $(x+5)$, width 3, perimeter 28 cm. a) Equation. b) Find x . c) Area.

7. (4 marks)

Regular pentagon, side $(2x-1)$ cm, perimeter 45 cm. a) Equation. b) Find x . c) Side length.

8. (5 marks)

a) Solve $3(2x - 1) = 2(x + 5) + 3$. b) Show $4(x + 2) = 2(2x - 1)$ has no solution and explain.